PLAYER DEMOGRAPHICS ANALYSIS AND VISUALIZATION OF INDIAN PREMIER LEAGUE (CRICKET)



Andre KURNIAWAN, Cong YAO, Gowri KHOTAKOTA, Lakshmimahesh SAHU, Naveen KOPPOLU

1. Introduction

1.1 Overview of the Project

- This project focuses on analyzing and visualizing data from the Indian Premier League (IPL). The aim is to uncover trends, evaluate player performance, and explore correlations between metrics.

1.2 Dataset Source

- The dataset was sourced from publicly available IPL data, including match statistics, player performance metrics, and team dynamics. Specific files include player data, match summaries, and delivery details.

- The Source via: https://www.kaggle.com/datasets/ramjidoolla/ipl-data-set

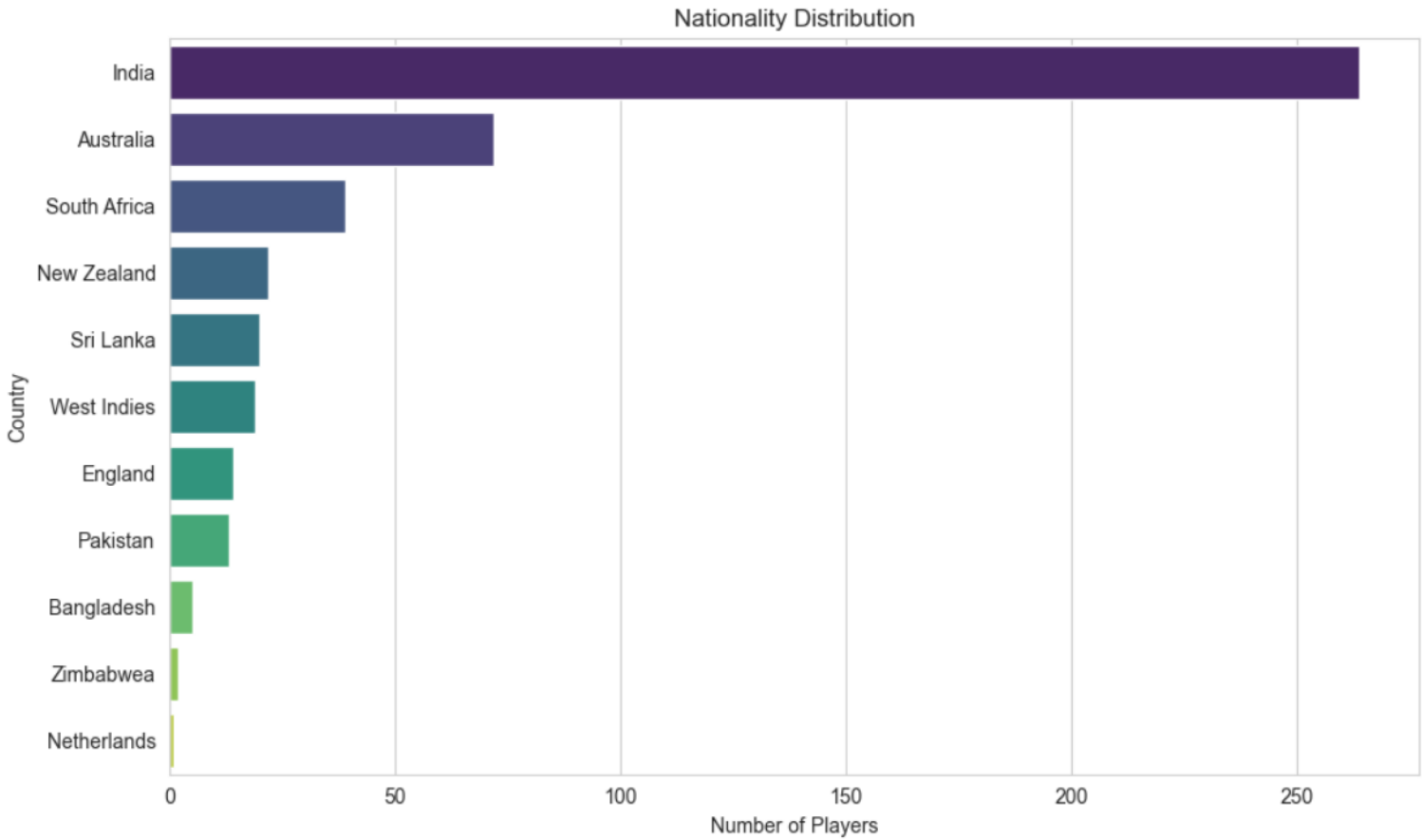
1.3 Objectives

- The project objectives include:
  - Performing demographic analysis of IPL players.
  - Analyzing player roles, performance trends, and consistency.
  - Building predictive models to estimate player performance.
  - Exploring team performance in home and away matches.
  - Identifying patterns using clustering techniques.

1.4 Team Roles

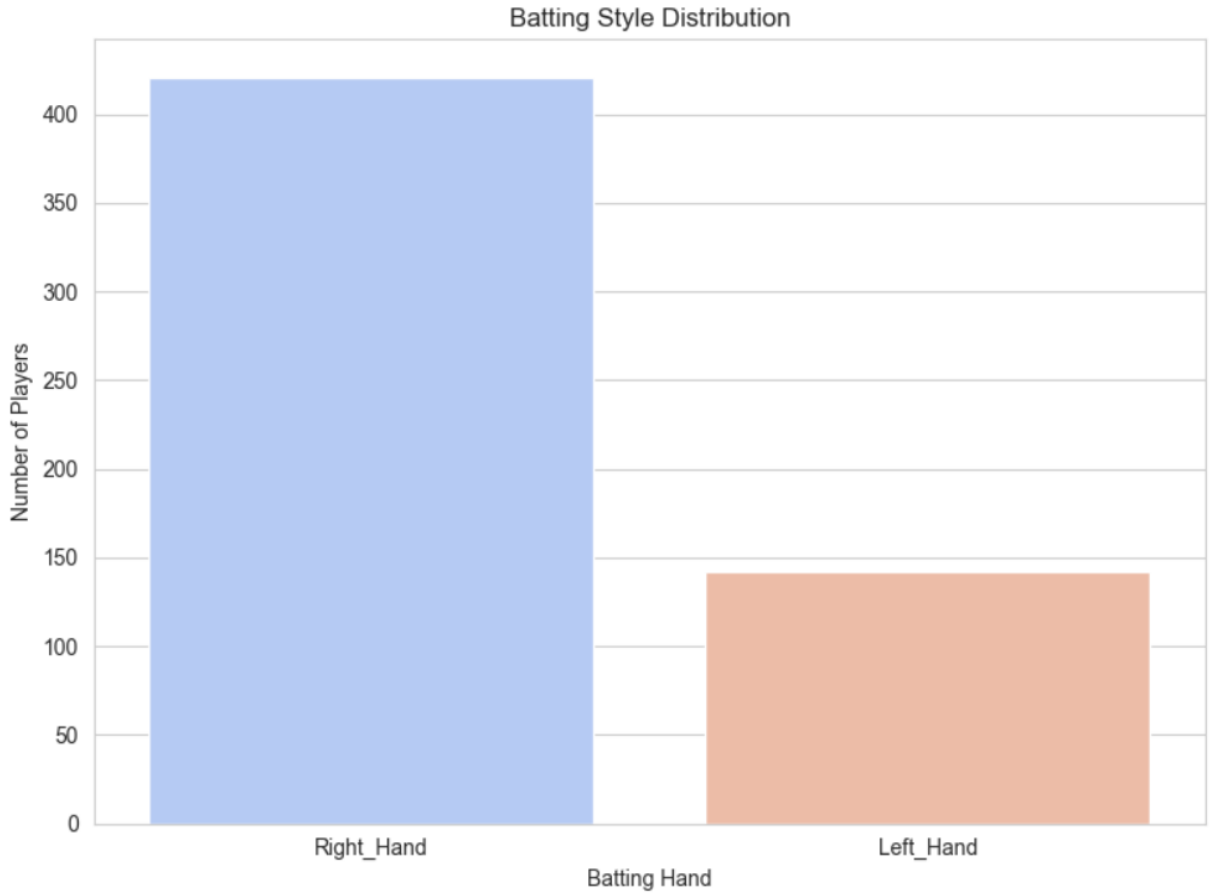
- - Naveen: Demographic Analysis, Player Role Analysis.
  - Cong: Player Trends and Patterns, Player Performance Prediction.
  - Gowri: Correlation Analysis, Player Trends and Patterns.
  - Mahesh: Diversity Analysis, Clustering Players Based on Preferences.
  - Andre: Data Visualization, Conclusion and Insights, Team Performance Analysis, Player Performance and Consistency Analysis

DEMOGRAPHIC ANALYSIS



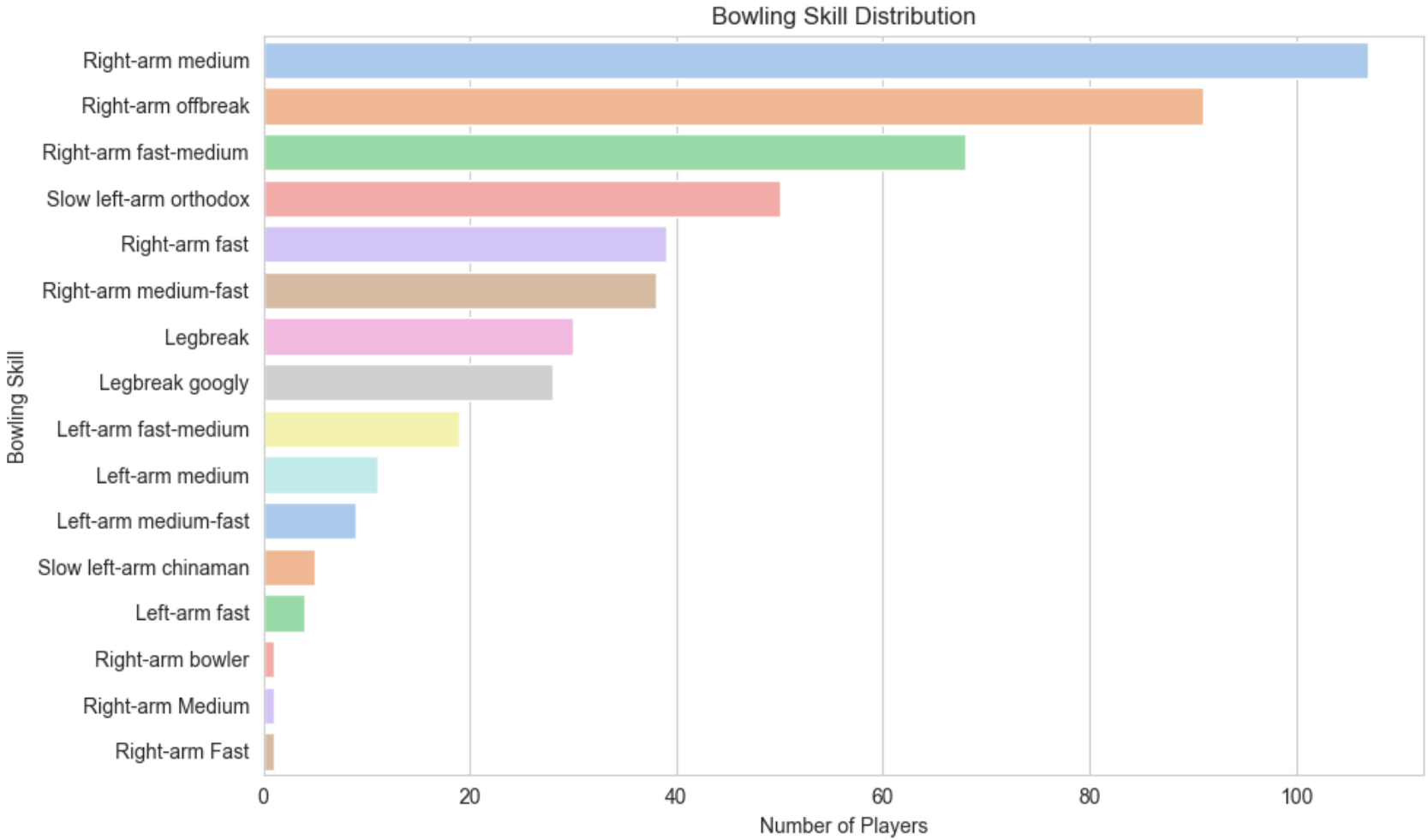
Country vs Number of Players

DEMOGRAPHIC ANALYSIS



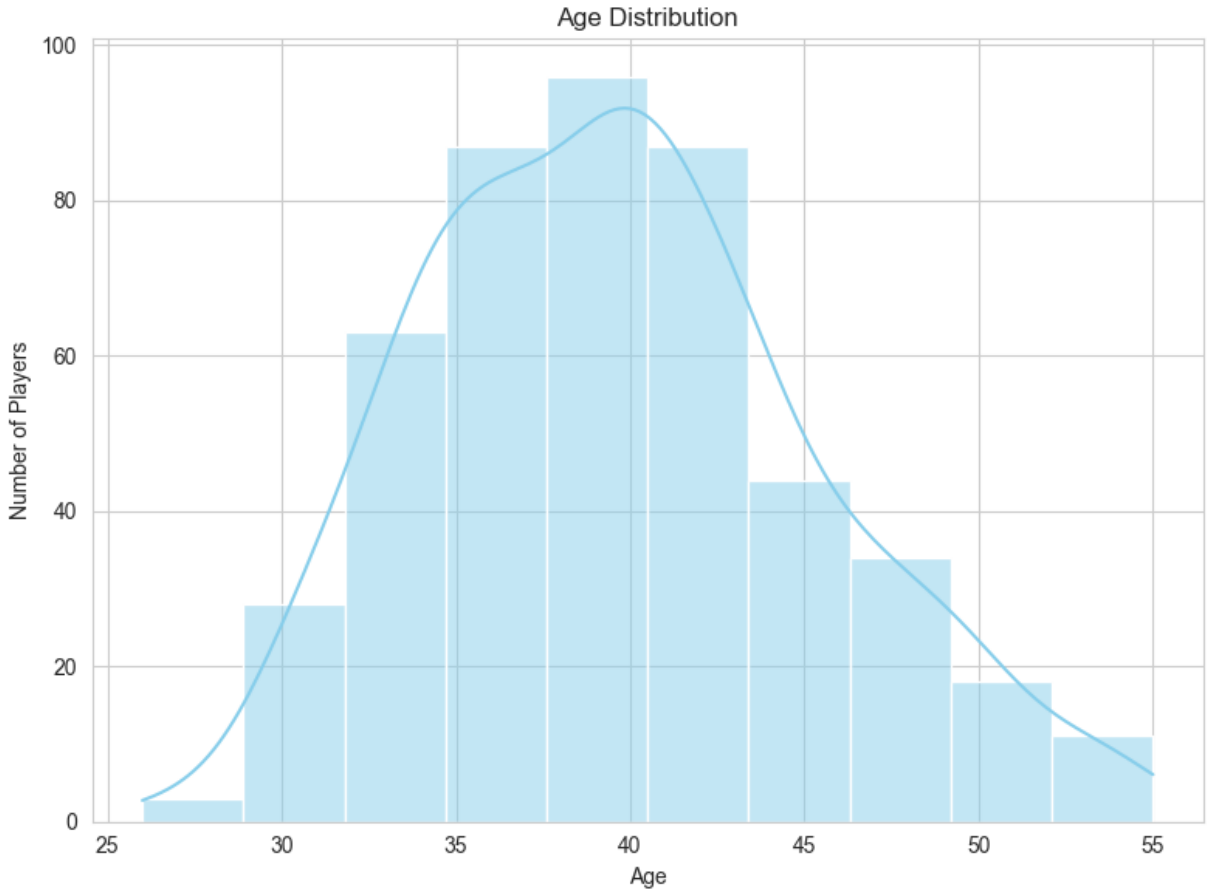
Number of Players vs Batting Hand

DEMOGRAPHIC ANALYSIS



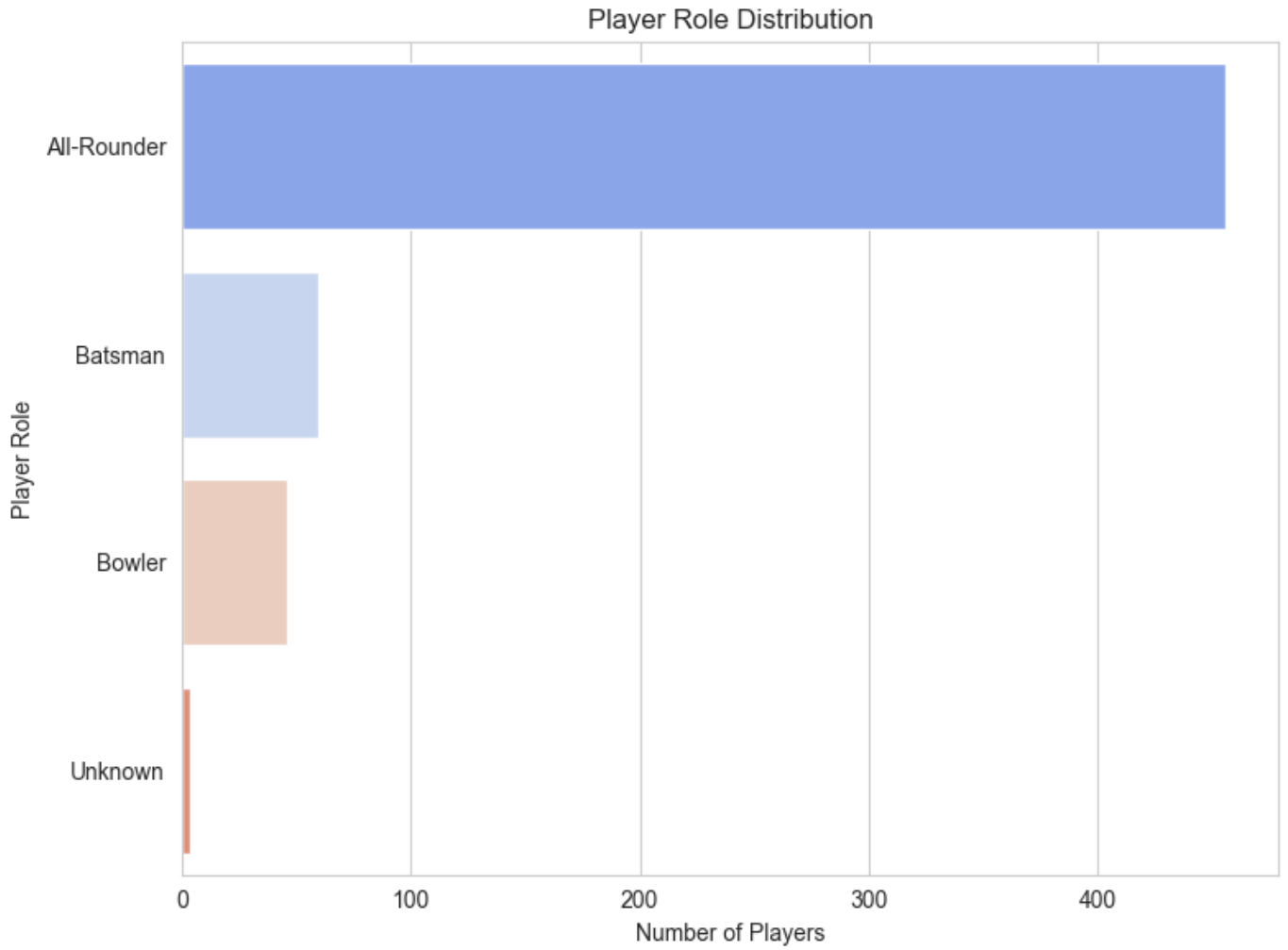
Bowling Skill vs Bowling Skill Distribution

DEMOGRAPHIC ANALYSIS



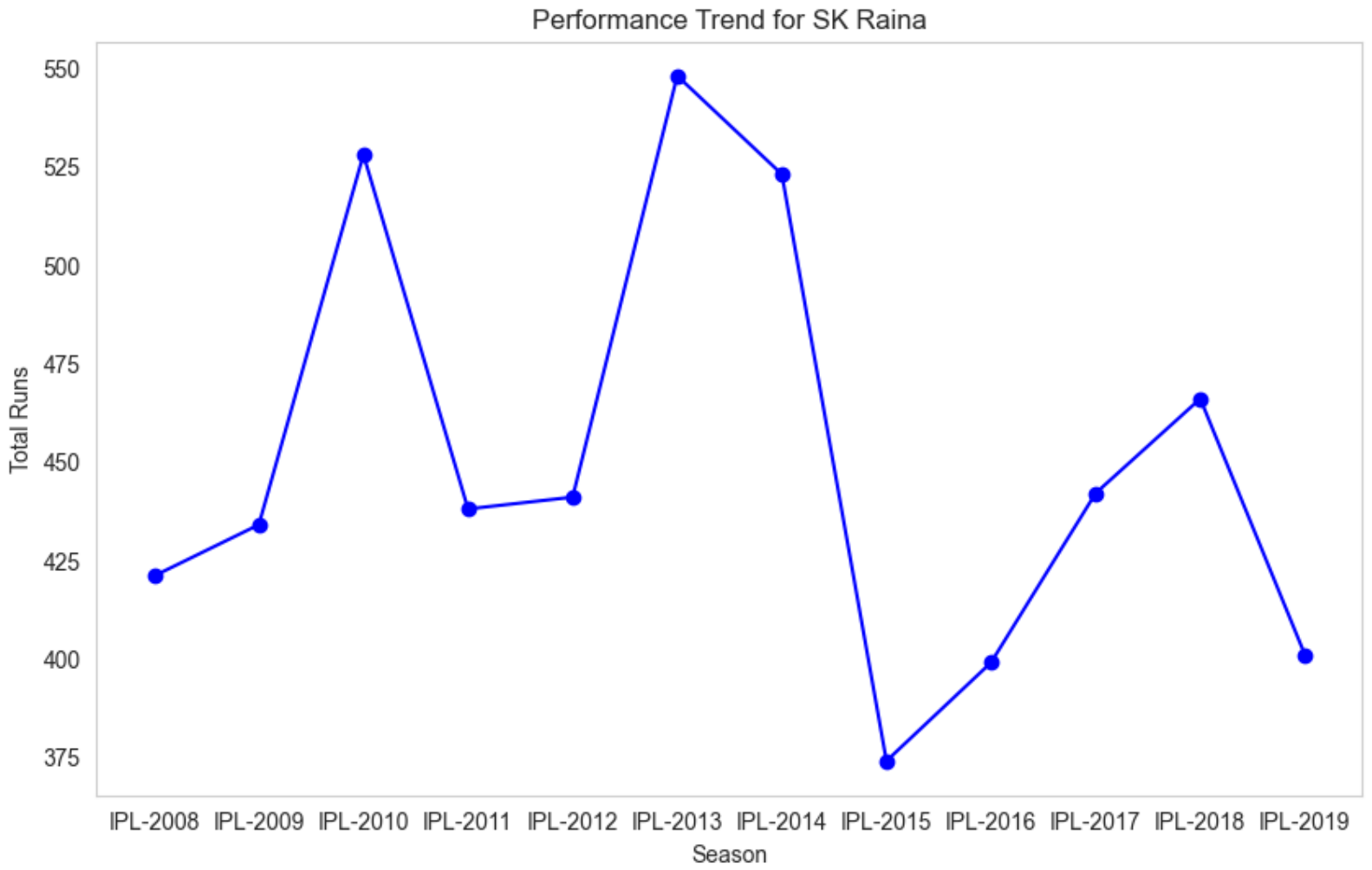
Number of Players vs Age

PLAYER ROLE ANALYSIS



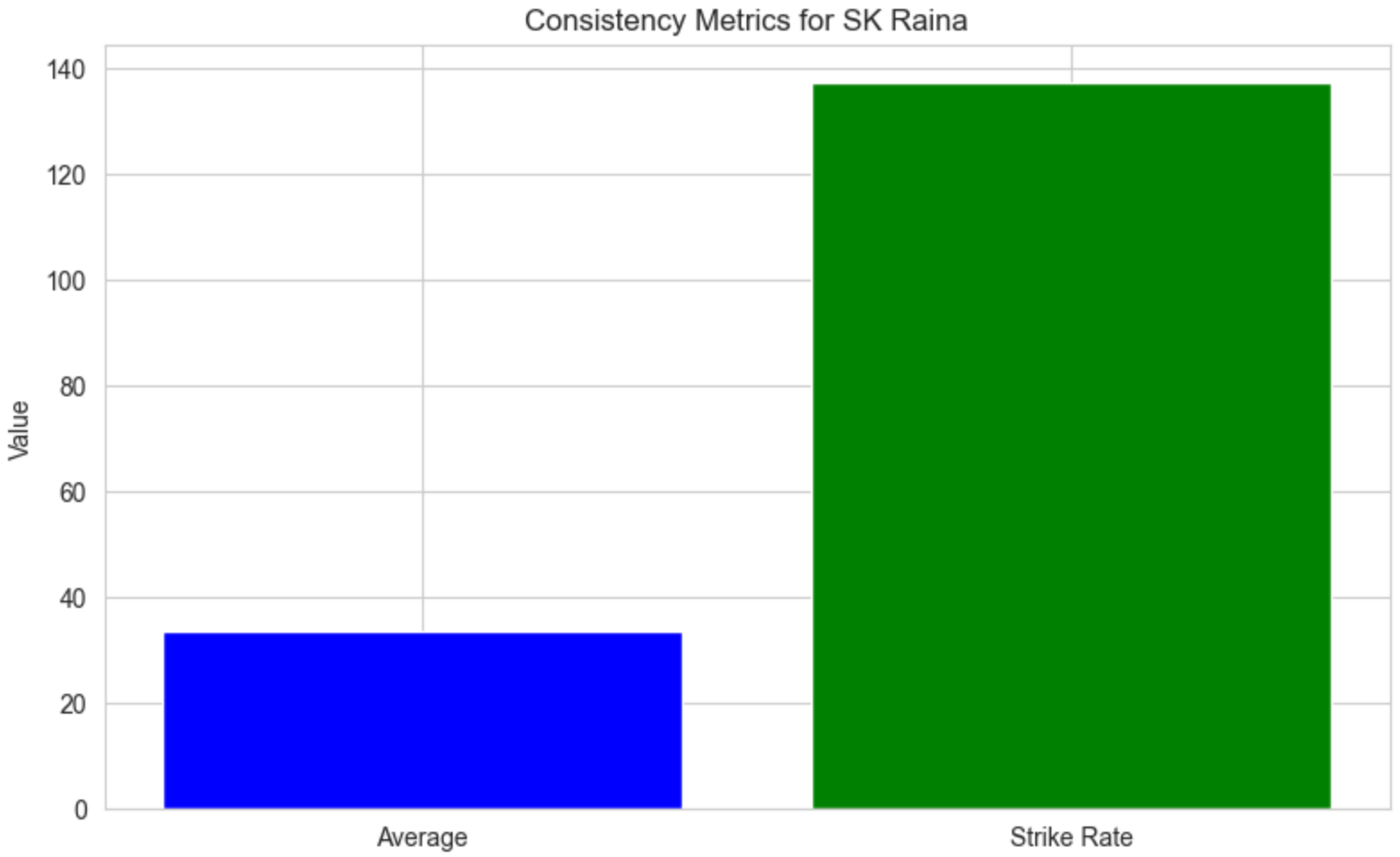
Player Role vs Player Role Distribution

PLAYER TRENDS AND PATTERN



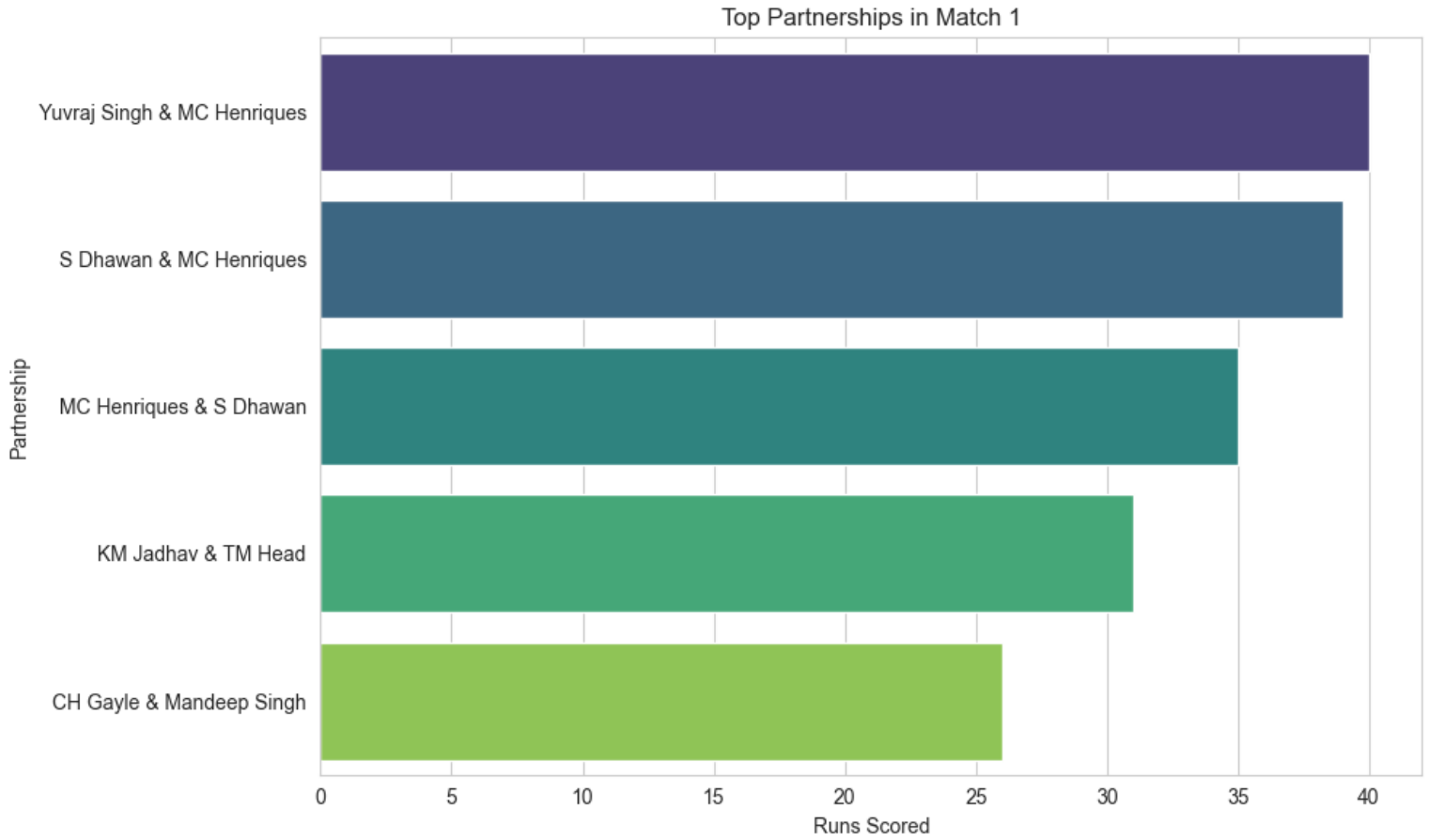
Total Runs vs Season

PLAYER TRENDS AND PATTERN



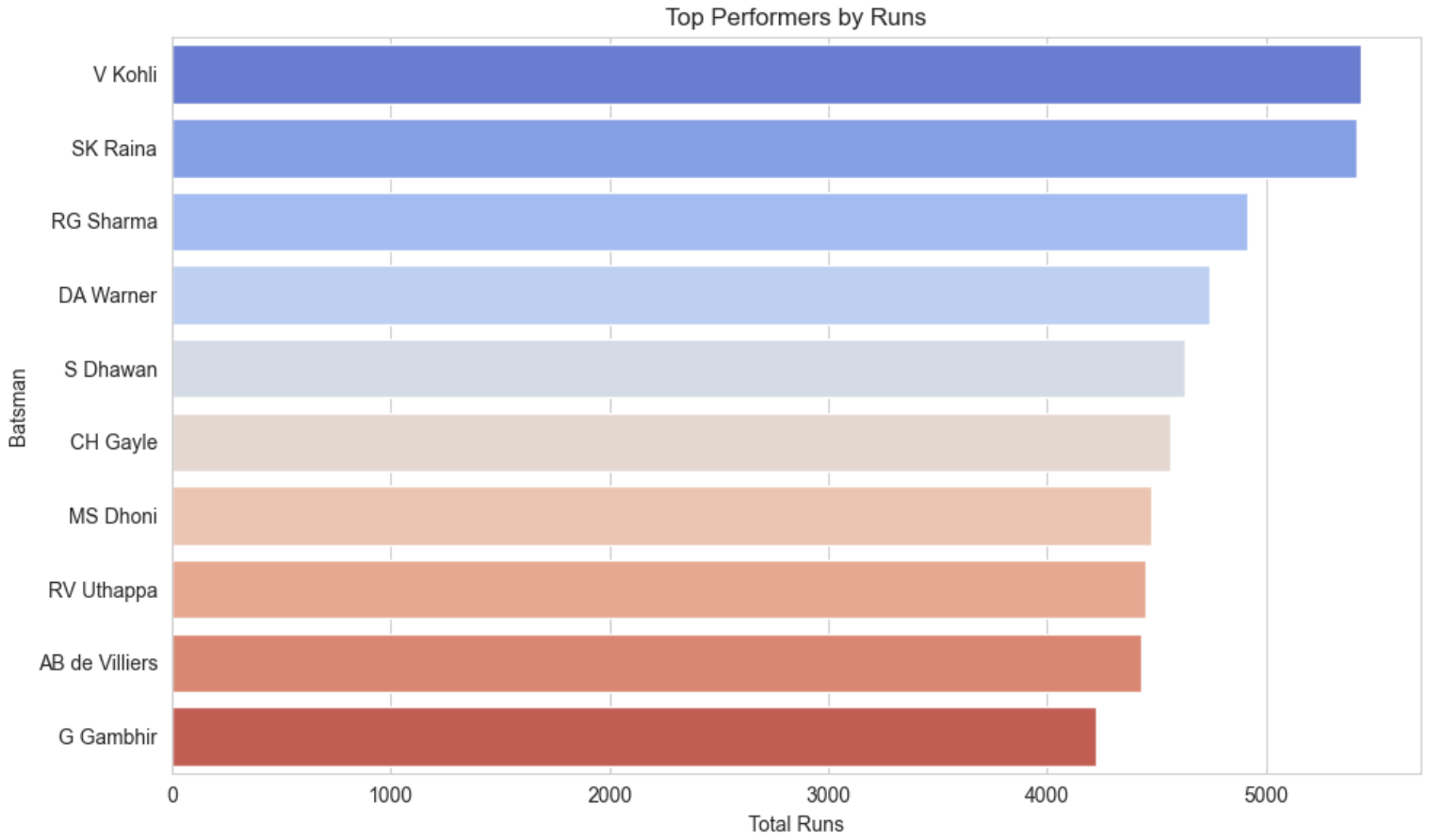
Value vs Average & Strike Rate

PLAYER TRENDS AND PATTERN



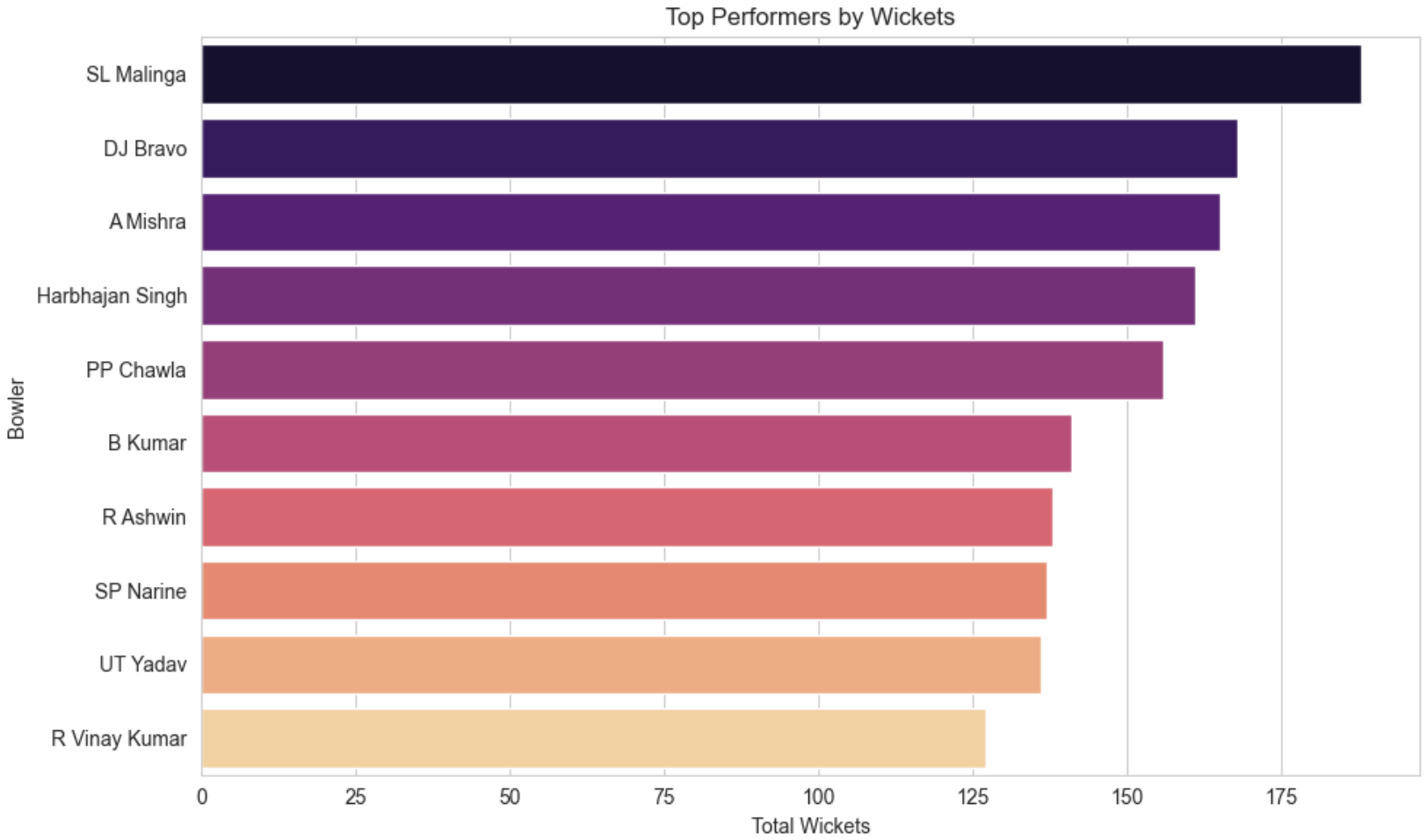
Partnership vs Runs Scored

PLAYER TRENDS AND PATTERN



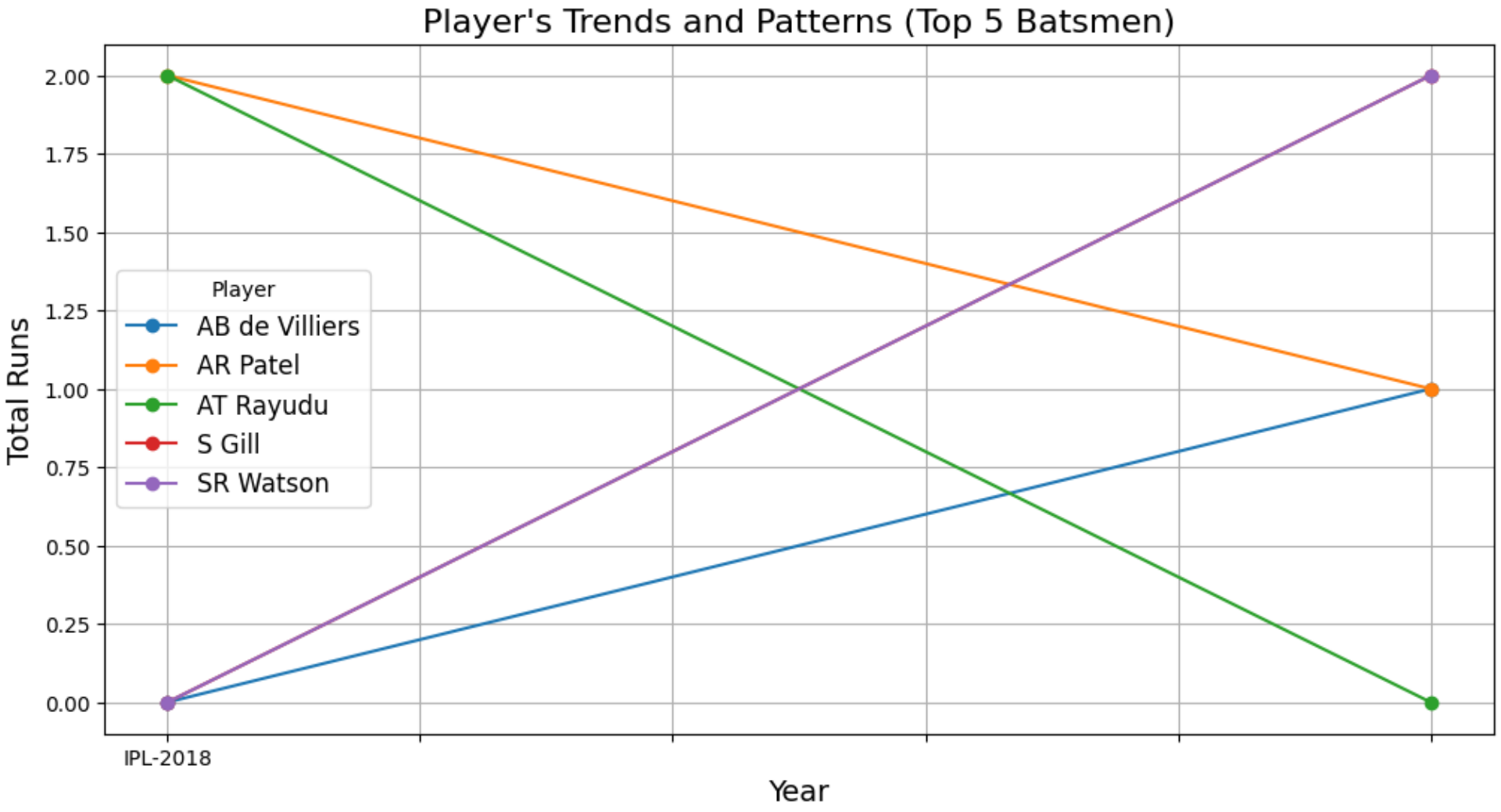
Batsman vs Total Runs

PLAYER TRENDS AND PATTERN



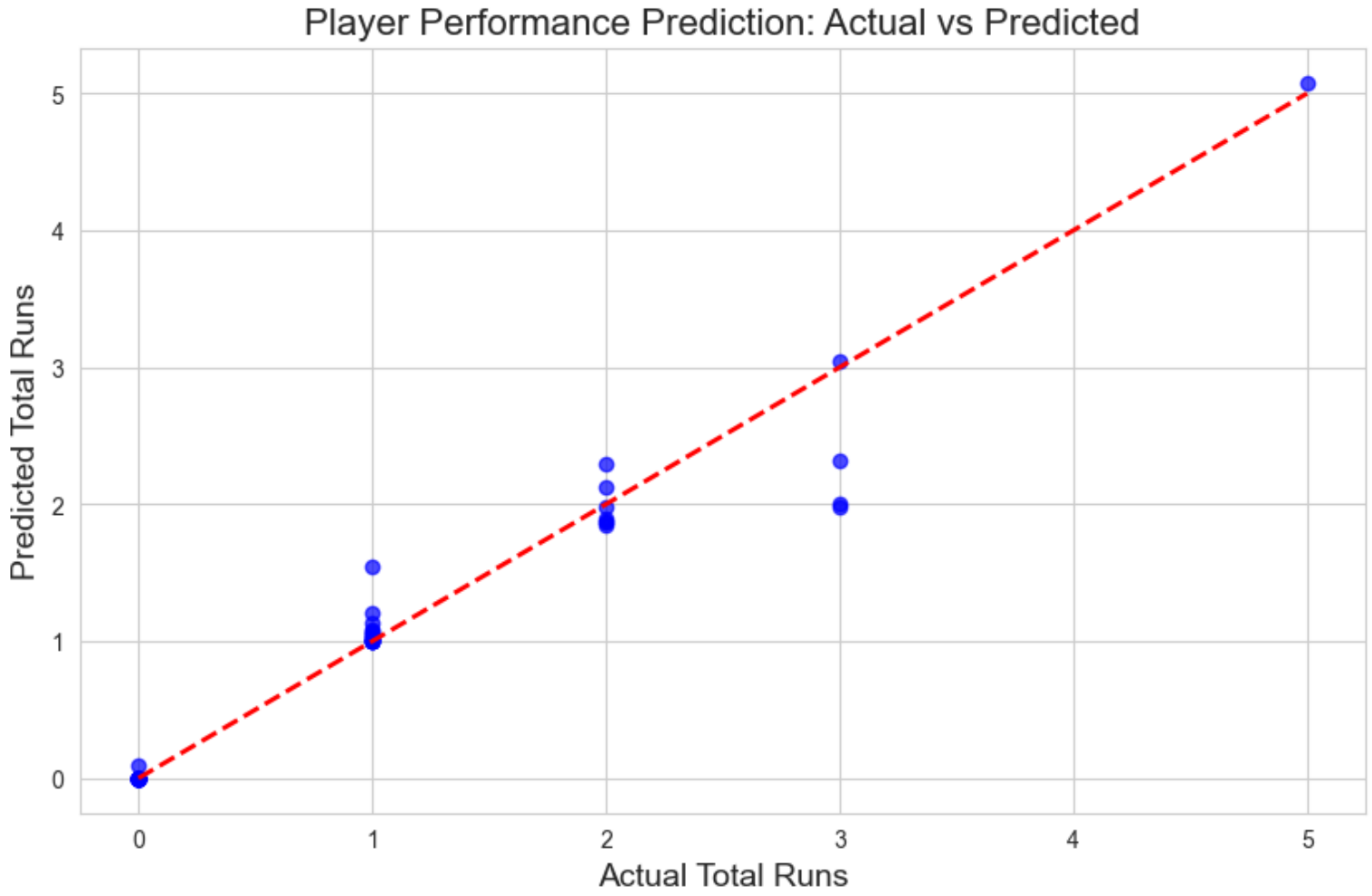
Bowler vs Total Wickets

PLAYER PERFORMANCE PREDICTION



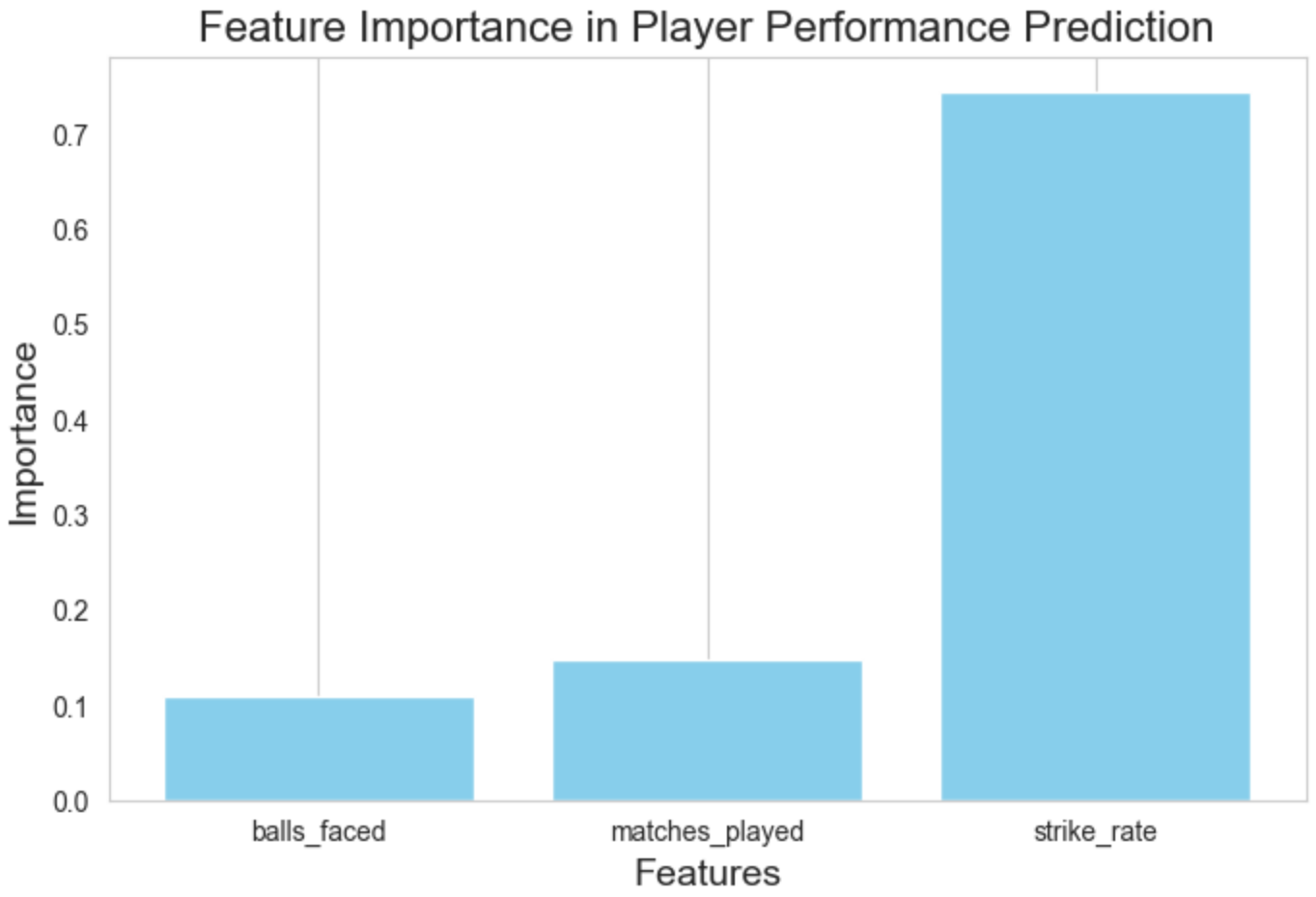
Total Runs vs Year

PLAYER PERFORMANCE PREDICTION



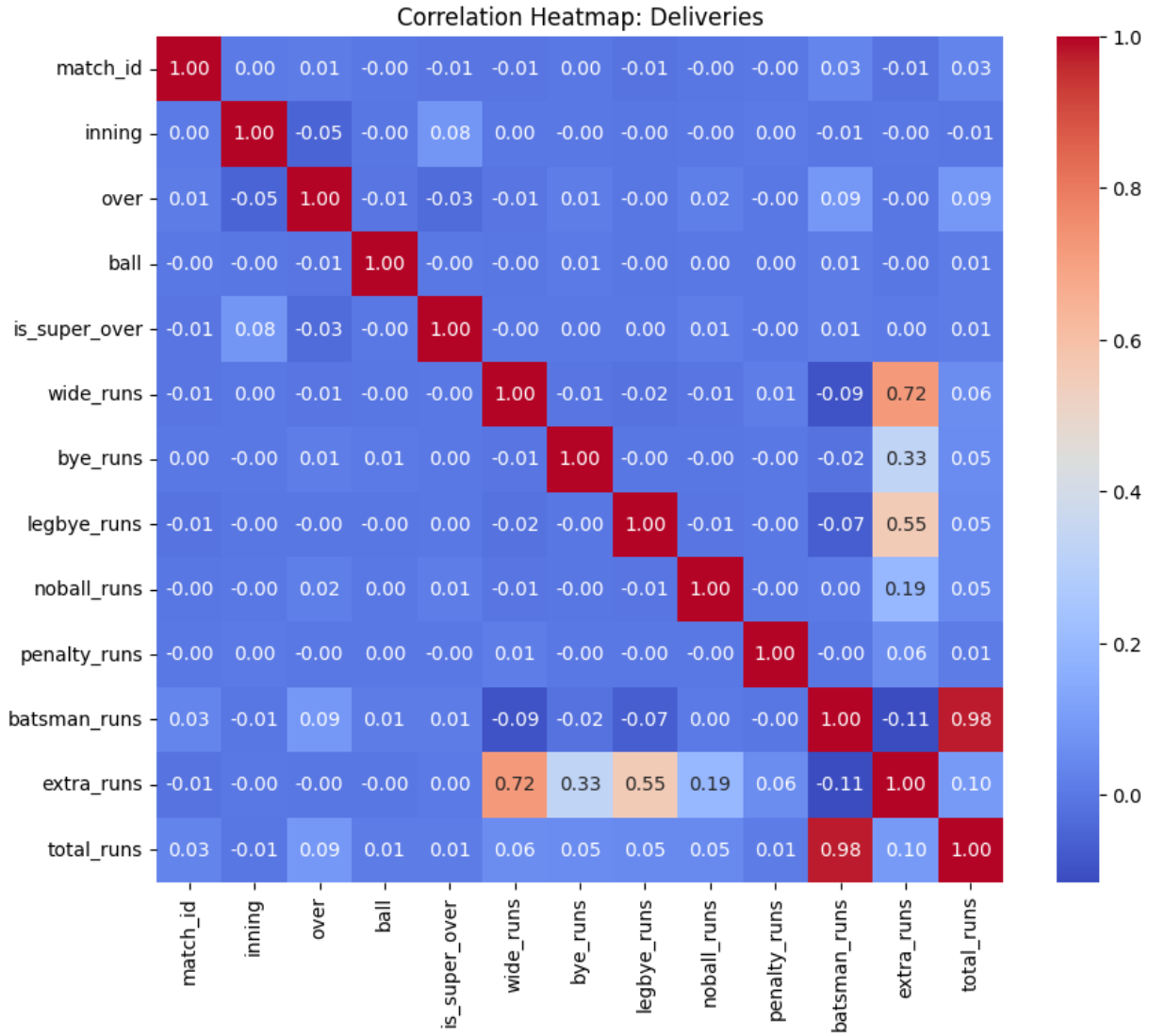
Predicted Total Runs vs Actual Total Runs

PLAYER PERFORMANCE PREDICTION

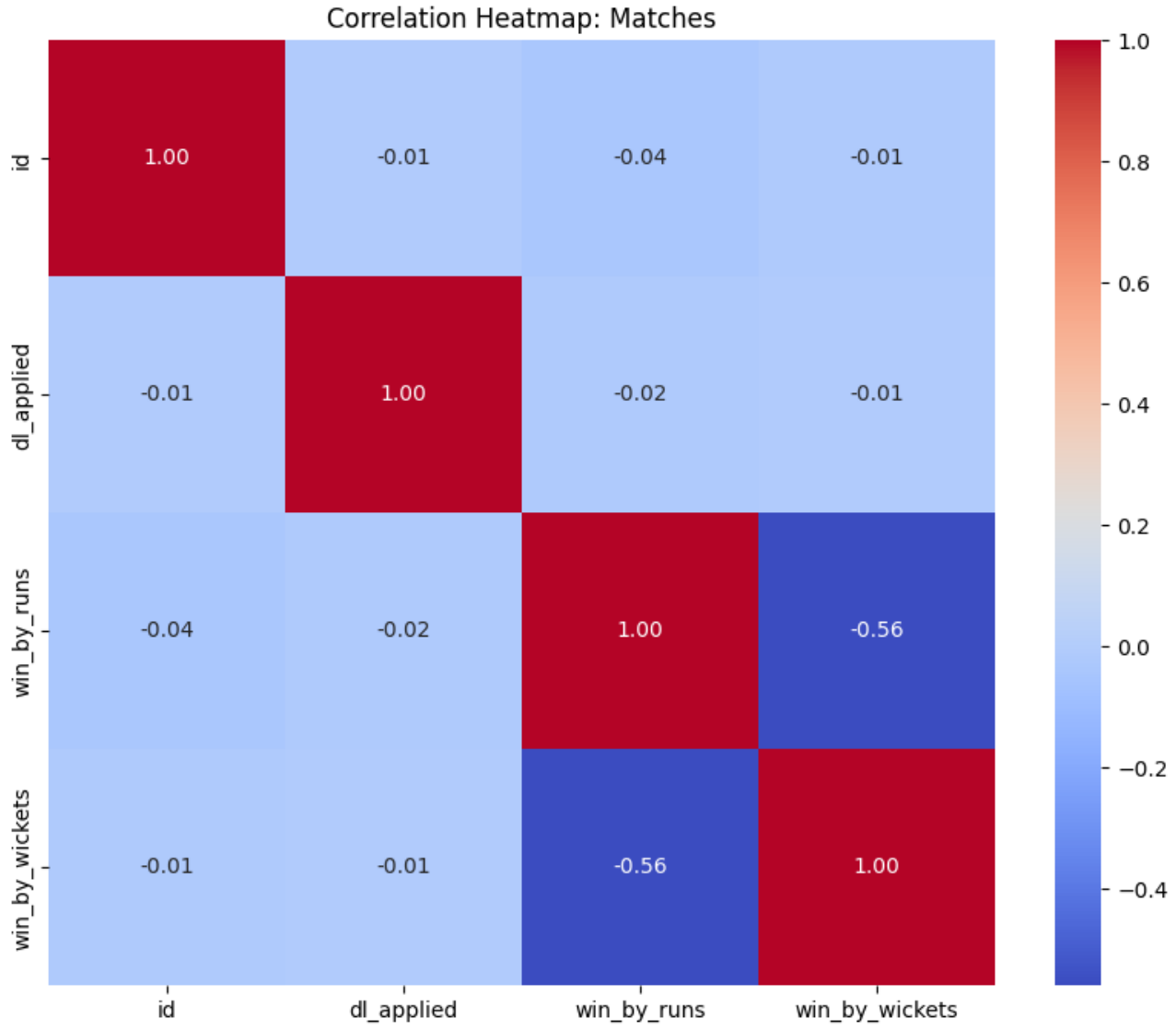


Importance vs Features

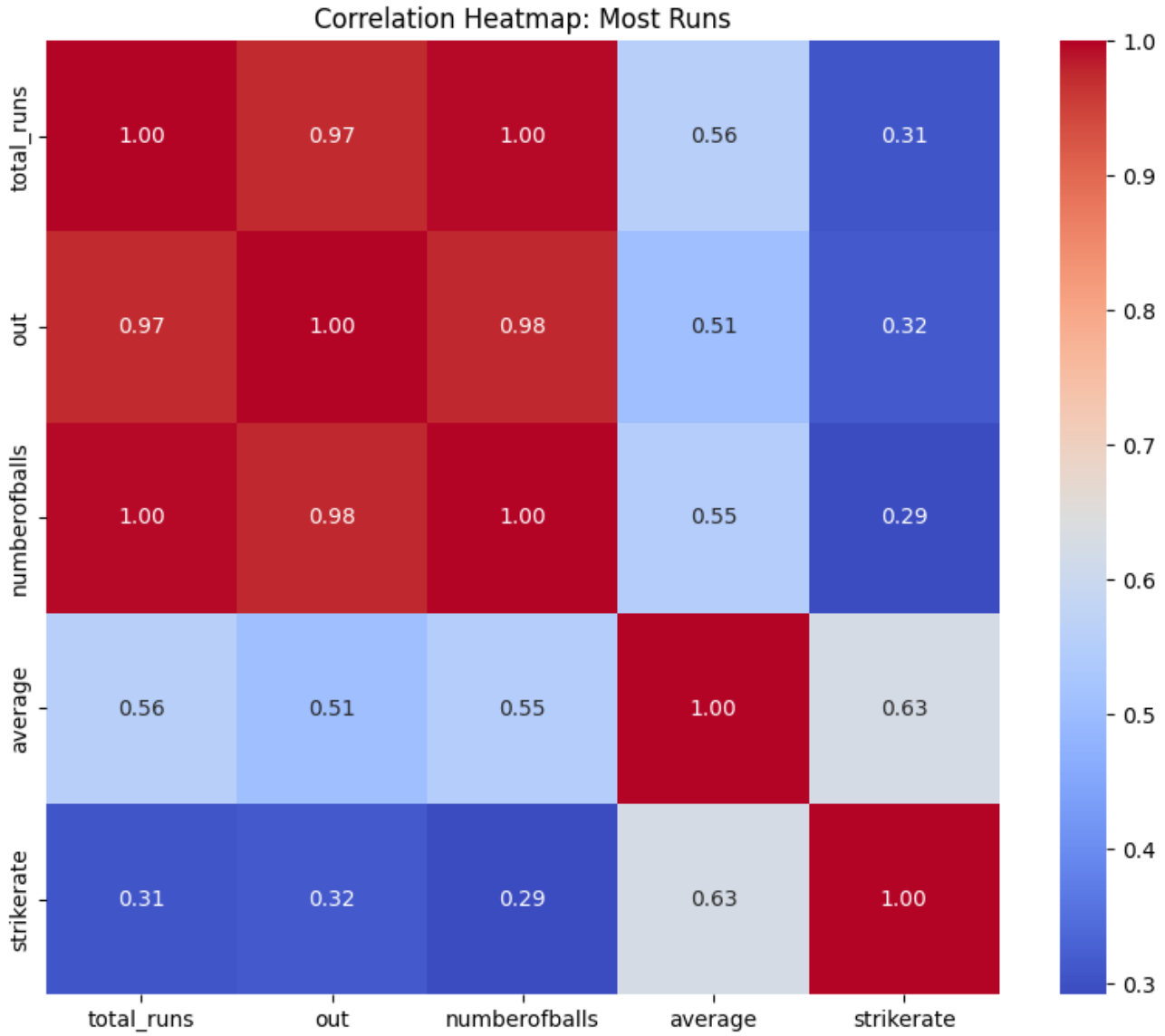
CORRELATION ANALYSIS



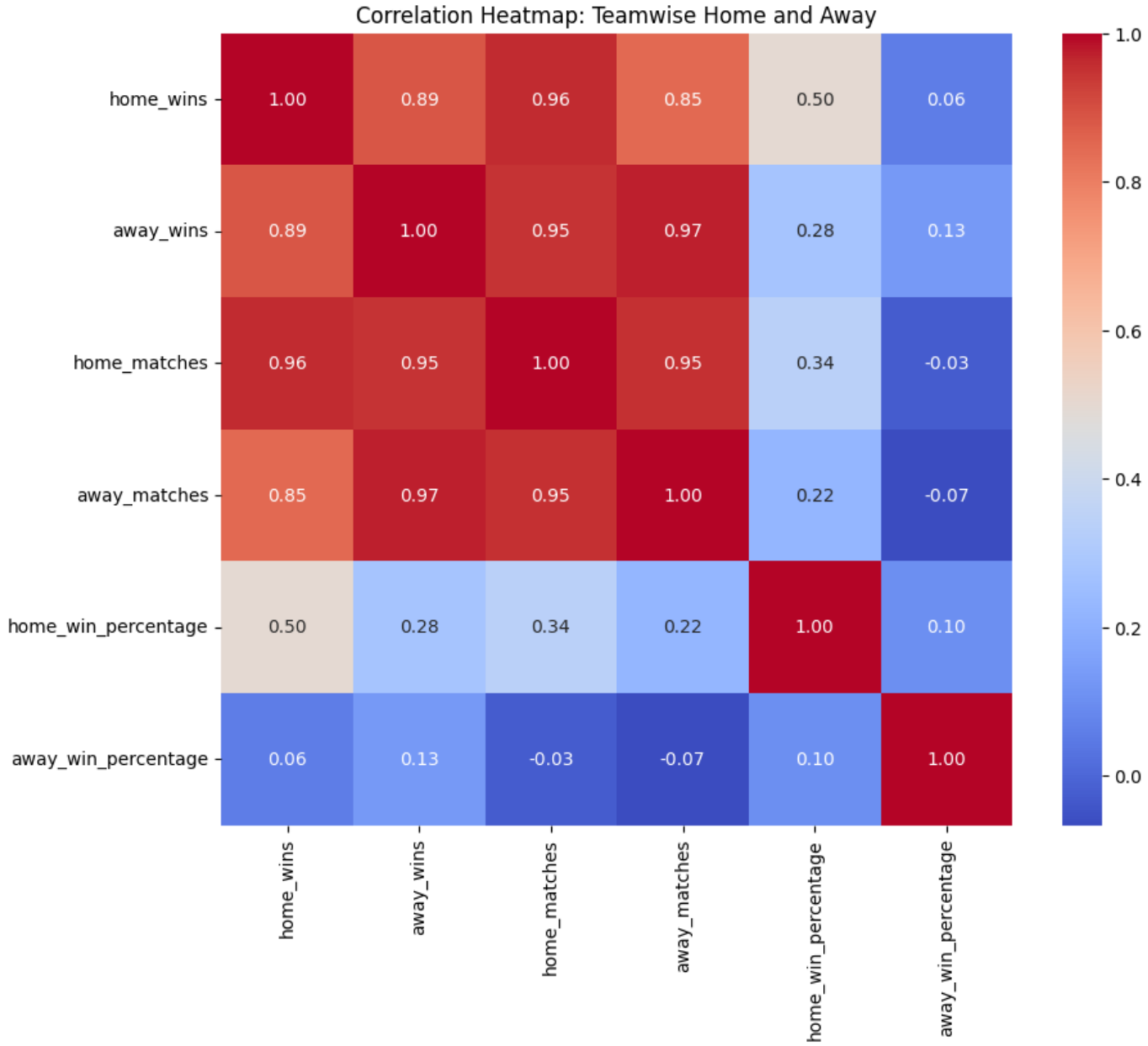
CORRELATION ANALYSIS



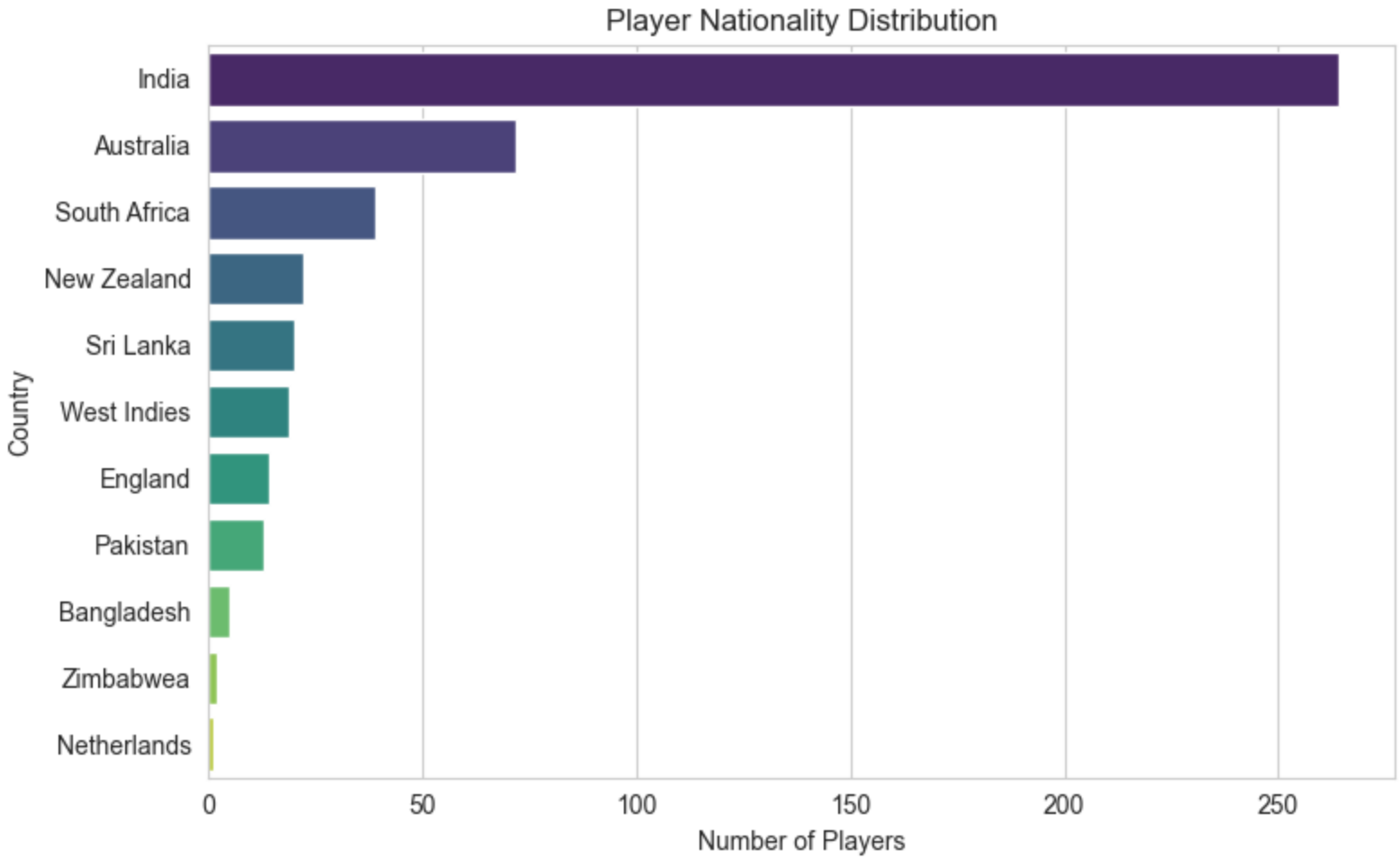
CORRELATION ANALYSIS



CORRELATION ANALYSIS

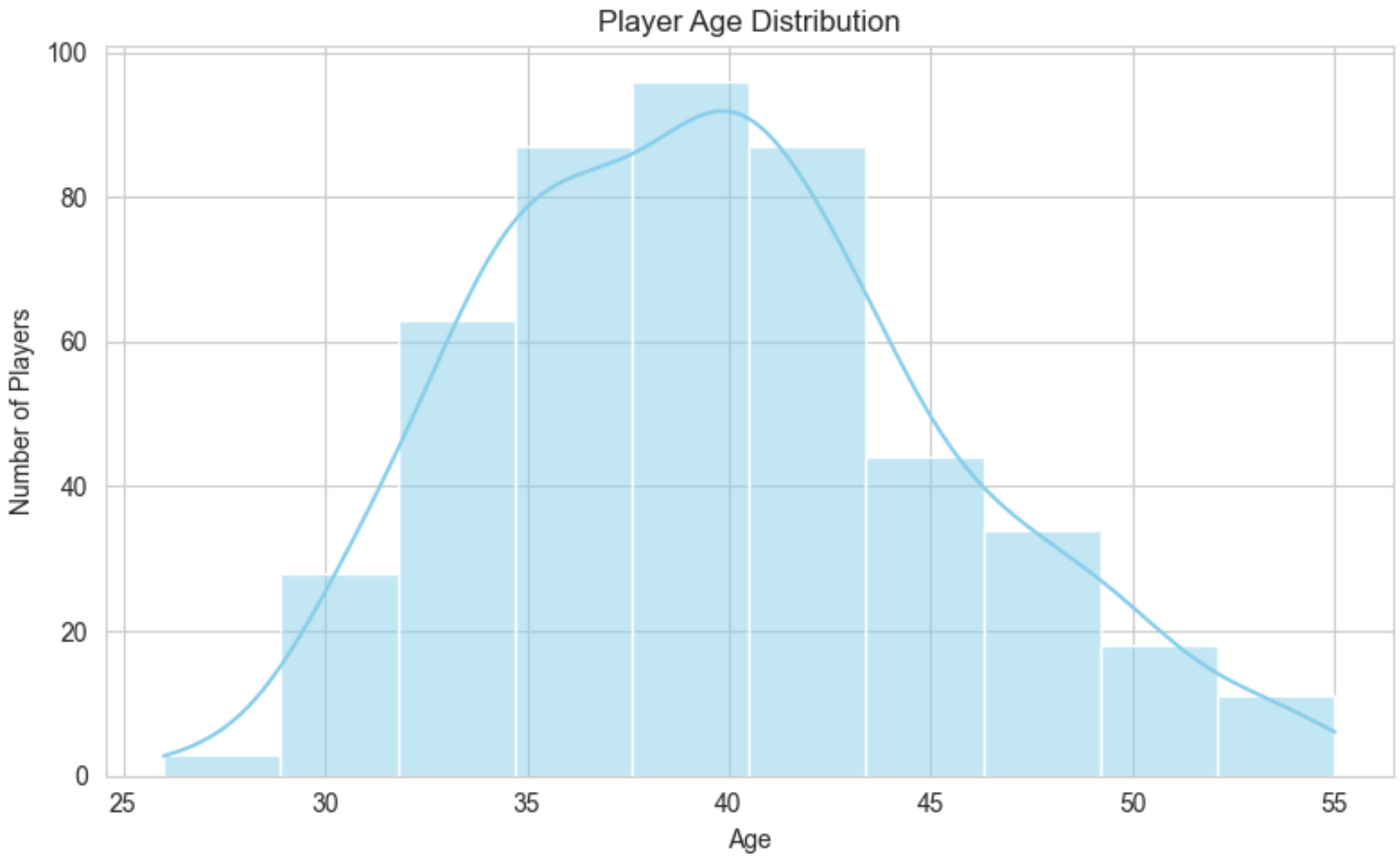


CORRELATION ANALYSIS



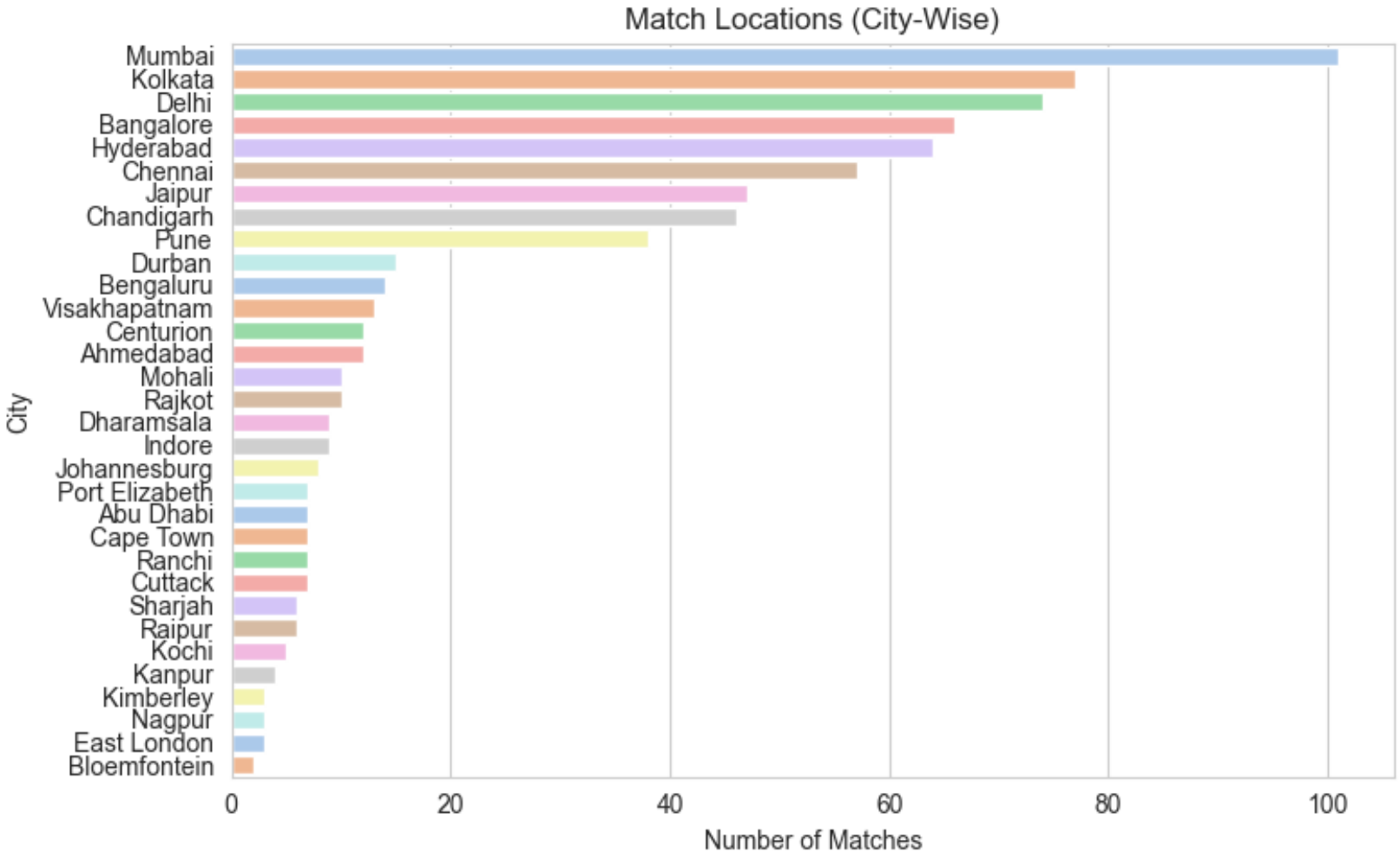
Country vs Numbers of Players

CORRELATION ANALYSIS



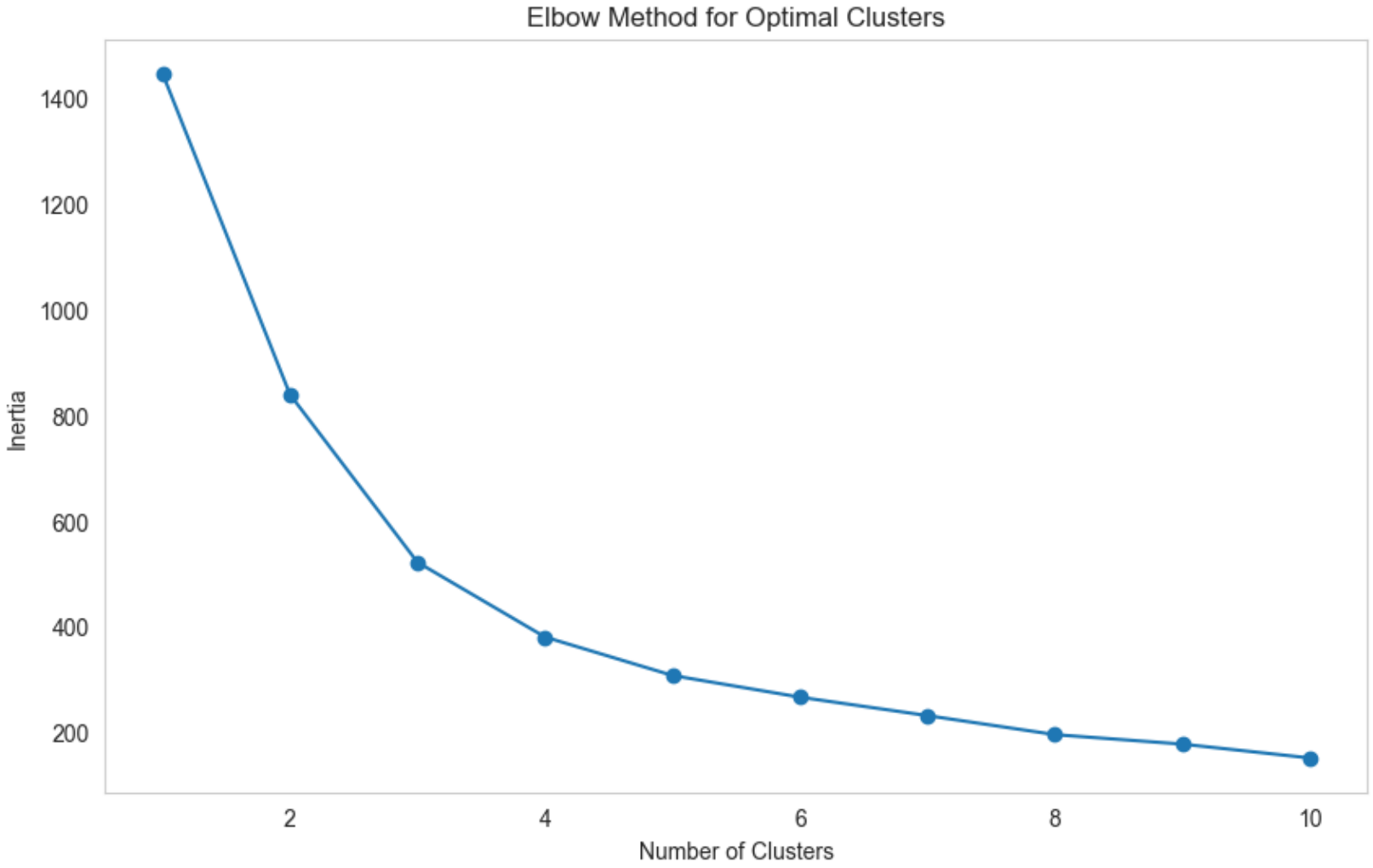
Number of Players vs Age

CORRELATION ANALYSIS



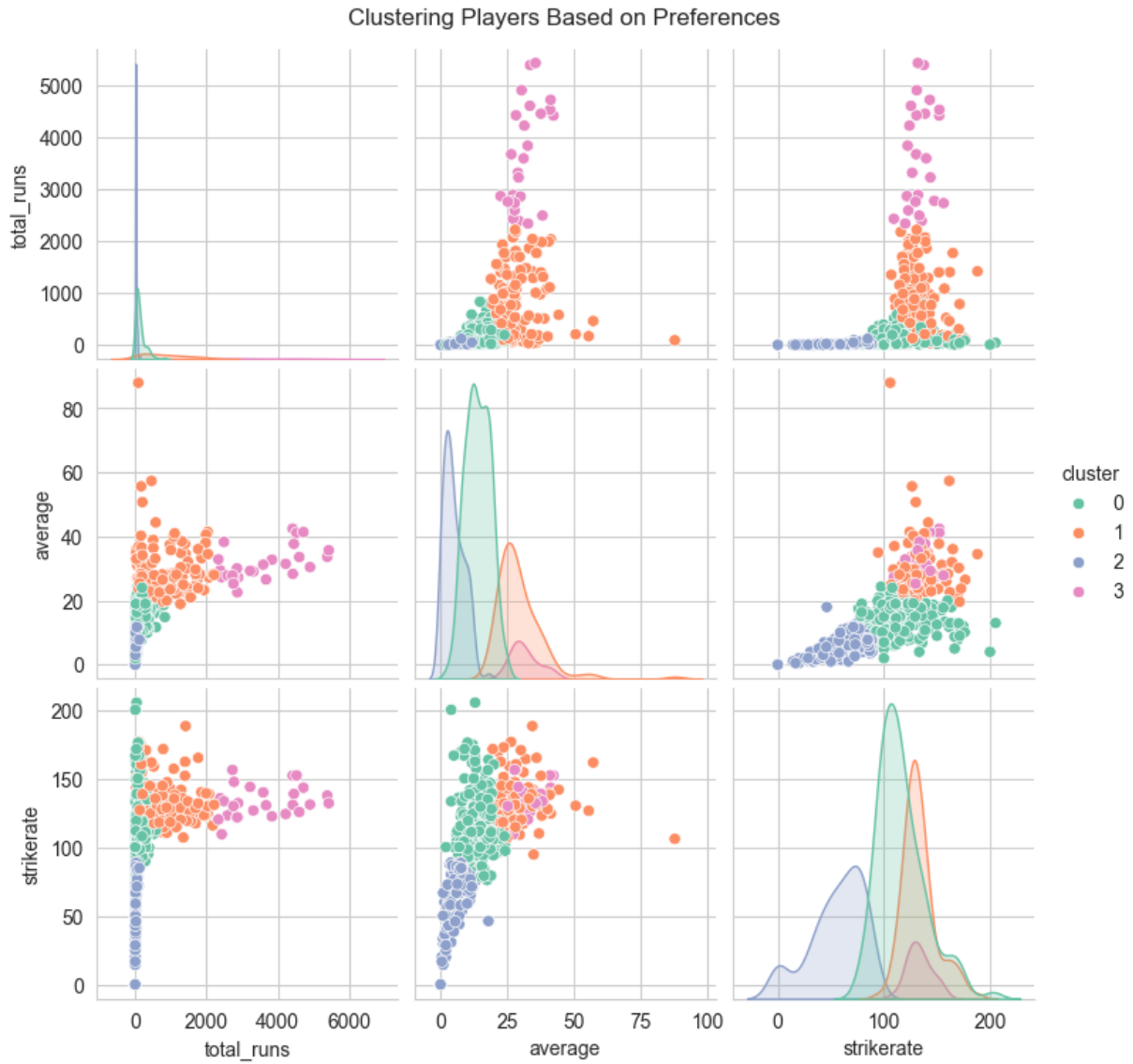
City vs Number of Matches

CLUSTERING PLAYER BASED ON PREFERENCES

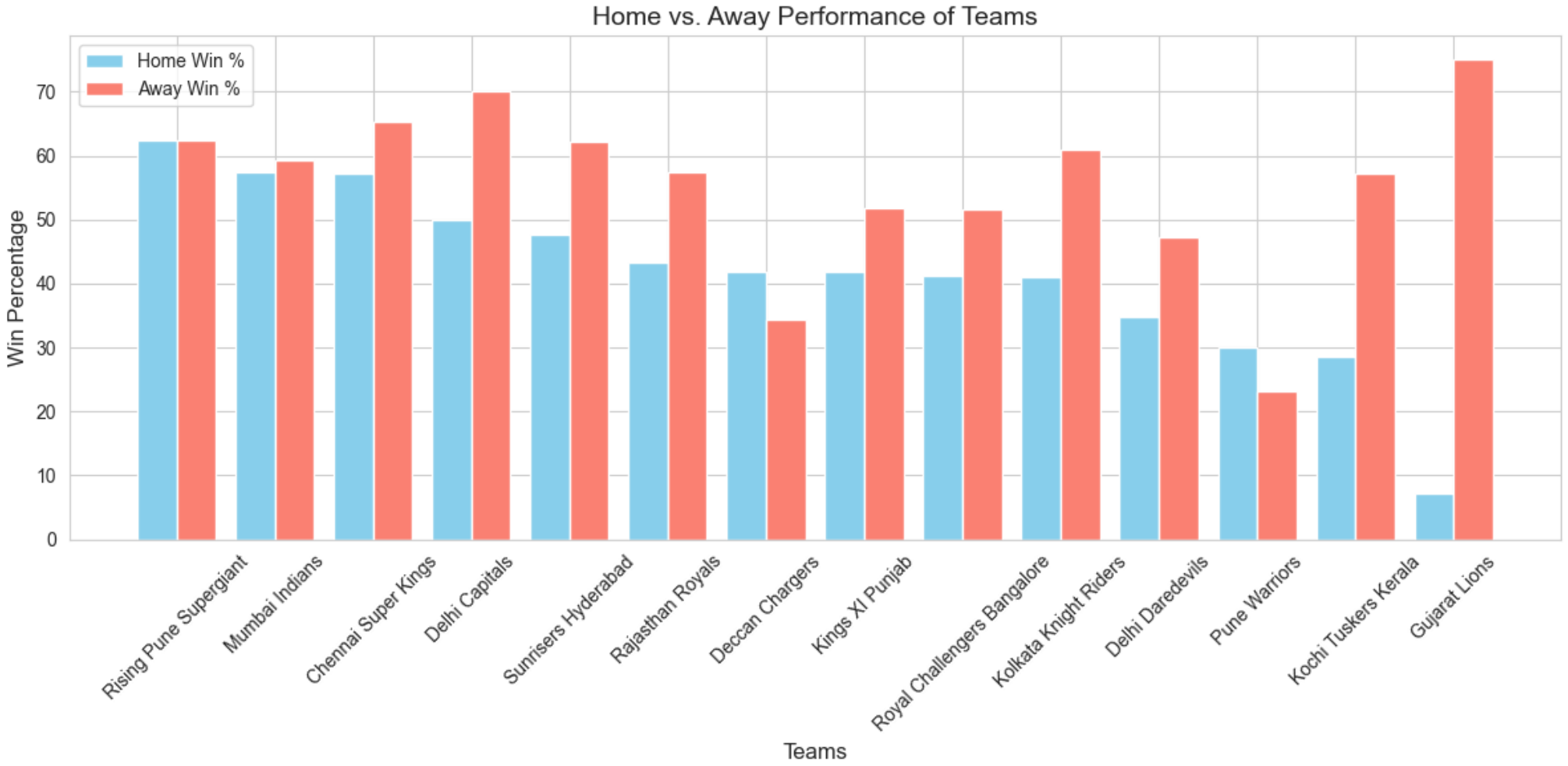


Inertia vs Number of Clusters

CLUSTERING PLAYER BASED ON PREFERENCES

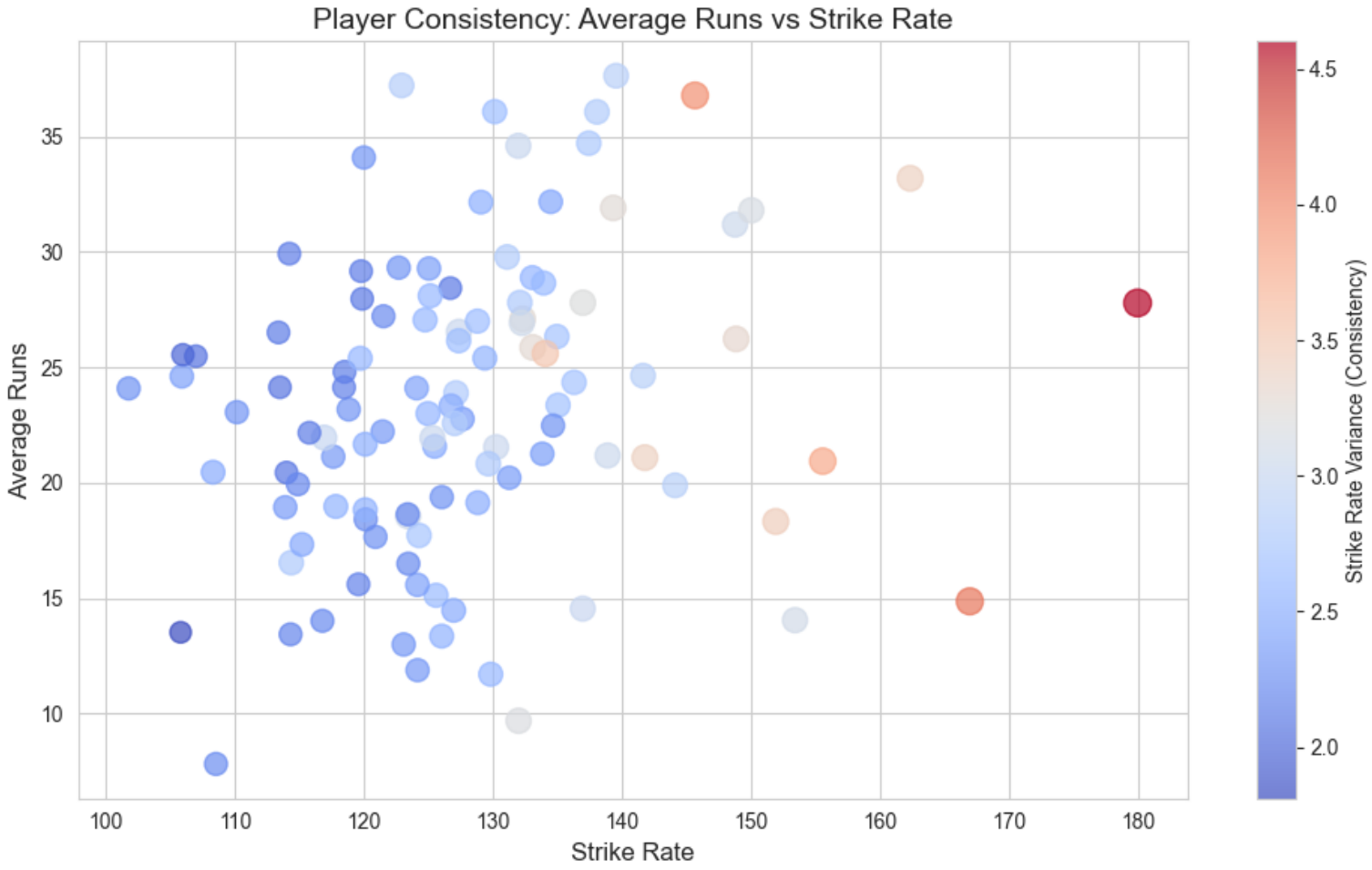


TEAM PERFORMANCE ANALYSIS: HOME VS. AWAY



Win Percentage Vs Teams

PLAYER PERFORMANCE AND CONSISTENCY ANALYSIS



Average Runs vs Strike Rate

Conclusion

- **Demographic Analysis:** This analysis offers a detailed view of player diversity, focusing on nationality, age, and playing styles. It highlights IPL's global reach and helps identify trends in the types of players that contribute most to team success.
- **Player Role Analysis**: By categorizing players into roles such as Batsman, Bowler, and All-Rounder, this analysis uncovers the balance within teams and shows how each role impacts team strategies and match outcomes.
- **Player Trends and Patterns:** This analysis tracks player performance over time, revealing trends and consistency. It also highlights key partnerships and standout players who make a significant impact on match results.
- **Player Performance Prediction:** Using historical data, this analysis helps predict future player performances, providing valuable insights into a player's potential contributions in upcoming matches.
- **Correlation Analysis:** This analysis explores the relationships between different performance metrics (e.g., runs vs. strike rate), offering a deeper understanding of how key factors influence player and team success.
- **Clustering Players Based on Preferences:** By clustering players with similar performance traits, this analysis allows teams to better understand player types, optimize their lineup, and strategize more effectively.
- **Team Performance Analysis (Home vs. Away):** Examining team performance at home versus away games, this analysis provides valuable insights into team strengths, weaknesses, and the importance of playing conditions, especially home-field advantages.
- **Player Performance and Consistency Analysis:** Focusing on consistency, this analysis evaluates player performance based on metrics like average runs, strike rate, and performance variance, identifying the most reliable players over time.

Overall Summary

These analyses offer a comprehensive understanding of IPL players and teams, revealing key insights into player performance, consistency, roles, and strategic elements. They provide valuable data to help teams improve performance, make informed decisions, and optimize strategies for better match outcomes.